Sheet 60 Thu 8 Apr. 2027 *Communicating a proof: implication, converse, equivalence*

1. Warm-up · Quick [DEM] State the converse of: "if a quadrilateral is a rhombus, then its diagonals are perpendicular".

   Ans. "If a quadrilateral has perpendicular diagonals, then it is a rhombus".

2. Core · True/false [DEM] True or false? Justify. The previous converse is true.

   Ans. False. Counter-example: a kite (not a rhombus) has perpendicular diagonals.

3. Core + · Write-up [DEM] Is the equivalence "$x^2 = 4 \Leftrightarrow x = 2$" correct? Correct it.

   Ans. No: $x = -2$ also works. We have the implication $x = 2 \Rightarrow x^2 = 4$ and the equivalence $x^2 = 4 \Leftrightarrow (x = 2 \text{ or } x = -2)$.

4. Review · Quick [CMP] Compare 2^{10} and 10^3.

   Ans. $2^{10} = 1\,024 > 1\,000 = 10^3$.

5. Thread · Quick [REP] Between which consecutive integers does $\dfrac{7}{4}$ lie on a number line? Give its decimal value.

   Ans. Between 1 and 2; 1.75.

Sheet 61 Mon 26 Apr. 2027 *Class discussion: comparing methods*

1. Warm-up · Quick [GEO] Distance between $A(-2;1)$ and $B(1;5)$?

   Ans. $\sqrt{3^2 + 4^2} = 5$.

2. Core · Method [GEO] Give three methods to prove that A, B, C are collinear.

   Ans. (i) $\det\left(\overrightarrow{AB}, \overrightarrow{AC}\right) = 0$; (ii) the equation of (AB) is satisfied by C; (iii) $AB + BC = AC$: equality of distances, equivalent to $B \in [AC]$.

3. Core + · Reflection [GEO] To prove that two lines given by $y = m_1 x + p_1$ and $y = m_2 x + p_2$ are parallel, which method is quickest?

   Ans. Compare the gradients: $m_1 = m_2$; strictly parallel if in addition $p_1 \neq p_2$.

4. Review · Factorise [DEV] Factorise $6x + 9$.

   Ans. $3(2x + 3)$.

5. Thread · Method [DIV] Write 360 as a product of prime factors. How many divisors does it have?

   Ans. $360 = 2^3 \times 3^2 \times 5$; $(3+1)(2+1)(1+1) = 24$ divisors.

Synthesis problems, reasoning and starter routines

Sheet 62 Tue 27 Apr. 2027 *Synthesis problems across several areas of the curriculum*

1. Warm-up · Quick [EQ] Solve $5x - 3 = 2x + 9$.

   Ans. $3x = 12$, $x = 4$.

2. Core · Problem [PRO] A price rises by 20 %, then a fixed discount of 1 000 Ar is applied, giving 11 000 Ar. What was the original price?

   Ans. $1.2p - 1\,000 = 11\,000$, $p = \dfrac{12\,000}{1.2} = 10\,000$ Ar.

3. Core + · Method [REG] $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the point of intersection of the curves, then solve $f(x) < g(x)$.

   Ans. $2x + 1 = -x + 7$: $x = 2$, point $(2;5)$; $f(x) < g(x) \Leftrightarrow 3x < 6 \Leftrightarrow x < 2$.

4. Review · Quick [QP] A recipe for 4 people needs 250 g of flour. How much for 6 people?

   Ans. $250 \div 4 \times 6 = 375$ g.

5. Thread · Quick [GEO] A shape of area 4 cm^2 is enlarged with scale factor 3. What is the area of the enlargement?

   Ans. Area $\times 3^2$: 36 cm^2.

   Pitfall: multiplying the area by 3 instead of 3^2.